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Surface recombination in the direct simulation Monte Carlo method

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This work is aimed at the development of surface chemistry models for the Direct Simulation Monte Carlo (DSMC) method applicable to non-equilibrium high-temperature flows about reentry vehicles. Probabilities of adsorption and Eley-Rideal recombination dependent on individual properties of each particular molecule and frequencies of desorption and Langmuir-Hinshelwood recombination are determined from macroscopic reaction rate data. Various macroscopic finite-rate surface reaction sets are used for the construction of the DSMC surface recombination models for the reaction cured glass and α-alumina surfaces. The models are implemented in the SMILE++ software system for DSMC computations, and detailed verification of the code is performed. The proposed approach is used to study the effects of surface recombination on the aerothermodynamics of a blunt body at high-altitude reentry conditions. *Published by AIP Publishing.* https://doi.org/10.1063/1.5048353

I. INTRODUCTION

At the early stages of the space vehicle aeroshell design, it is necessary to accurately predict the aerothermodynamic characteristics of the vehicle along its reentry trajectory (see, e.g., Ref. 1). A typical feature of the flow around the reentry vehicle is a significant degree of thermochemical non-equilibrium. Experimental modeling of high-altitude non-equilibrium flight environment in ground-based facilities is extremely difficult; therefore, numerical modeling is the main tool for studying such flows. There are two high-fidelity approaches to numerical simulation of gas flows: continuum and kinetic. The continuum approach based on the Navier-Stokes equations is inapplicable at high flight altitudes where the mean free path of molecules is comparable with the size of the vehicle, and the kinetic approach should be used for modeling such flows. The best-developed and efficient implementation of the kinetic approach is the Direct Simulation Monte Carlo (DSMC) method (Ref. 2).

The magnitude of the heat flux onto the reentry vehicle surface is largely determined by physical and chemical processes in the shock layer and on the surface of the thermal protection system (TPS). Reactions of dissociation of free-stream molecules into atoms and simpler molecules proceed in the high-temperature viscous shock layer on the spacecraft surface, which leads to cooling of the flow. Simultaneously, however, the spacecraft surface serves as a catalyst for exothermic reverse reactions of recombination, which can severalfold increase the heat flux to the vehicle (Refs. 3 and 4).

Much attention has been always paid to the modeling of gas-phase chemical processes in computational fluid dynamics (CFD) within the framework of both the continuum and kinetic approaches. Codes based on the continuum approach (e.g., Refs. 5–11) usually involve chemical reaction models that take into account the reaction rate dependence on translational and vibrational temperature (the model developed by Park, Ref. 12, is used most frequently). The most popular model for real engineering applications of the DSMC method is the total collision energy model (TCE, Refs. 13 and 2), where data on the macroscopic reaction rate under thermal equilibrium are used for determining the probability of the chemical reaction between two molecules (see also the recent modification of the model in Refs. 14 and 15). This phenomenological approach was generalized in order to take into account the effects of vibration dissociation favoring (see, e.g., Refs. 16–23). The main advantage of the phenomenological approach to modeling chemical reactions is its relative simplicity, possibility of easy extension to complex multispecies mixtures, and compatibility of such models with macroscopic models used in continuum CFD. There are many papers dealing with the validation of DSMC gas-phase chemistry models against data of ground-based and flight experiments (see, e.g., Refs. 16 and 24–32).

It should be noted that surface chemical processes at the atmospheric reentry conditions have been studied to a much smaller extent than gas-phase reactions, and the degree of uncertainty associated with modeling these processes is much higher. In continuum gas dynamics, surface processes can be taken into account with different degrees of detail: from a simple variant of a constant recombination probability (the value of this parameter equal to zero and unity correspond to the limiting cases of noncatalytic and perfectly catalytic walls, respectively) to sufficiently complex finite-rate surface chemistry models, which take into account the kinetics of various physical and chemical processes on the spacecraft surface. A number of models of surface chemical reactions with different levels of detail and accuracy have been proposed in the framework of the continuum approach (Refs. 4 and 33–40). These models include those based on empirical data as well as some modern models that employ the results of *ab initio* calculations.

As for the kinetic approach, and particularly the DSMC method, very little attention has been paid to the particle modeling of spacecraft aerothermodynamics taking into account

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surface catalytic processes. There are few publications dealing with this topic (Refs. 41–46); moreover, the influence of heterogeneous chemical reactions on high-altitude aerothermodynamics of space vehicles has not been studied by the DSMC method almost at all.

The principal goal of the present study is to provide researchers and engineers who use the DSMC method with an approach to modeling surface catalytic processes, which could be used in real engineering applications and could simultaneously ensure the level of detail equal to or greater than the conventional DSMC models of gas-phase chemical reactions. This paper describes activities that continue and extend the research started in Refs. 47 and 48. The main idea is to construct DSMC surface chemistry models based on detailed finite-rate macroscopic surface reaction sets. All the DSMC model parameters that determine the microscopic probabilities of the processes are obtained from the macroscopic reaction rate coefficients. In a certain sense, this approach is an extension of that used in the TCE model to surface reactions. This approach to surface chemistry modeling was partially developed in Ref. 43. A pioneering general method of constructing DSMC surface chemistry models on the basis of an almost arbitrary macroscopic finite-rate surface reaction set including various surface processes (adsorption, desorption, Eley-Rideal and Langmuir-Hinshelwood recombination reactions, etc.) is proposed in this paper.

The paper is structured as follows. Section II provides the necessary basic information on heterogeneous catalytic processes. The mathematical approach to obtaining microscopic parameters of DSMC surface chemistry models on the basis of prescribed macroscopic reaction rate coefficients for each type of the surface process separately is formulated in Sec. III, where general issues of implementation of the resultant models in the DSMC code are briefly described as well. Section IV describes some examples of various materials used for the thermal protection of space vehicles for which finite-rate macroscopic surface reaction sets in air are available in the literature; DSMC surface chemistry models are developed for these materials on the basis of the proposed approach. Section V deals with verification of the developed DSMC surface chemistry models implemented into the SMILE++ software system (Refs. 49 and 50) for numerical studies of aerothermodynamic problems on multiprocessor computers, developed in Russia at the Khristianovich Institute of Theoretical and Applied Mechanics (ITAM), which is an upgraded modification of the well-known SMILE system (Ref. 51). An example of the flow around a blunt body under atmospheric reentry conditions calculated with the use of the developed models of surface catalytic processes is considered at the end of the paper.

II. HETEROGENEOUS PROCESSES

A necessary stage for heterogeneous catalysis is adsorption of particles (atoms or molecules) on surface sites (it can be various crystalline lattice defects). Adsorbed particles can spontaneously leave the surface (the so-called desorption) and react with each other or with gas-phase particles colliding with the surface. As the result of these processes, the surface coverage is changed and new particles are produced.

TABLE I. Elementary surface processes. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

Impact mechanisms	
Adsorption	$A + S \rightarrow A_S$
Eley-Rideal recombination	$A_S + B \rightarrow AB + S$
Dissociative adsorption	$AB + S \rightarrow A_S + B$
Surface mechanisms	
Desorption	$A_S \rightarrow A + S$
Langmuir-Hinshelwood recombination	$A_S + B_S \rightarrow AB + 2S$

Elementary processes typical of heterogeneous catalysis are listed in Table I. A and B denote atoms or molecules, S is the surface site where adsorption occurs, and the subscript $_S$ denotes the particles adsorbed by the surface. All surface processes can be divided into two groups: *impact mechanisms*, which involve gas-phase particles colliding with the surface, and *surface mechanisms*, which involve only particles adsorbed by the surface. These processes are illustrated in Fig. 1.

Heterogeneous processes in the macroscopic description are characterized by the reaction rate coefficients $K(T)$. The dependence of these coefficients on the temperature T is usually assumed to be in the Arrhenius form

$$K(T) = AT^b \exp\left(-\frac{E_a}{kT}\right), \quad (1)$$

where k is the Boltzmann constant and E_a is the reaction energy barrier. These coefficients are different for different surface materials.

Heterogeneous processes in continuum CFD (e.g., based on Navier-Stokes equations) are usually taken into account through boundary conditions using the catalytic recombination coefficient γ, also referred to as the probability of recombination. It is equal to the fraction of all particles incident onto the surface and participating in heterogeneous recombination reactions, i.e.,

$$\gamma = \frac{J^{reac}}{J}, \quad (2)$$

where J is the total flux of particles onto the surface and J^{reac} is the flux of particles incident onto the surface and taking part in recombination reactions, i.e., in the Eley-Rideal or Langmuir-Hinshelwood recombination. This coefficient can be divided into two parts defining the Eley-Rideal recombination coefficient γ_{ER} and the Langmuir-Hinshelwood recombination coefficient γ_{LH}, respectively (clearly, $\gamma = \gamma_{ER} + \gamma_{LH}$). Under certain assumptions, γ can be determined from the reaction rate constants (1) of all processes involved in heterogeneous catalysis.

Note that there are various options of reaction schemes for heterogeneous catalytic processes, and only one example is presented in Table I. In other schemes, some of these processes can be neglected or included in a slightly different form; for example, in the Langmuir-Hinshelwood or Eley-Rideal

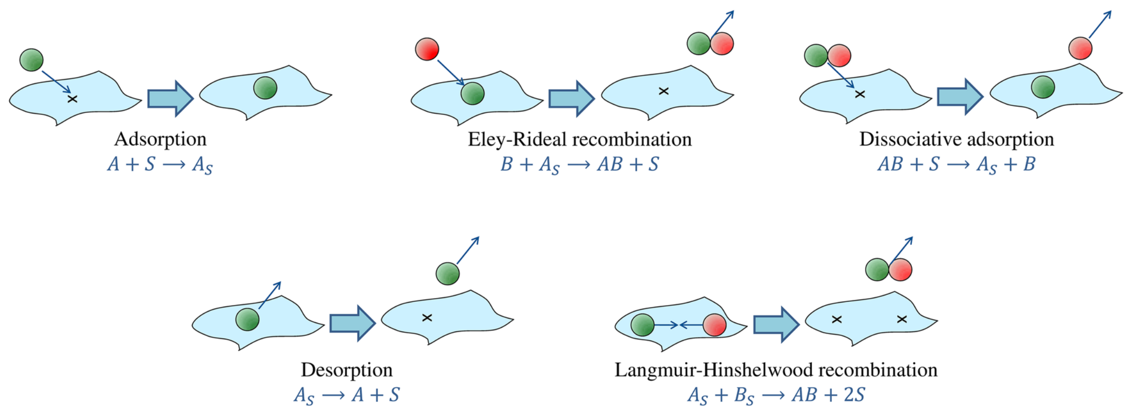


FIG. 1. Surface processes.

recombination, the resulting molecule AB can remain adsorbed on the surface after the reaction (i.e., AB_S); another possible variant is the adsorption of both resulting atoms (i.e., a reaction reverse to the Langmuir-Hinshelwood recombination is included) in the case of dissociative adsorption.

III. APPROACH TO THE DEVELOPMENT OF DSMC MODELS

In sets of macroscopic surface reaction, it is necessary to specify the list of elementary surface processes and the corresponding reaction rate coefficients. In the present work, we propose to obtain all the parameters required for the DSMC surface chemistry modeling (such as probabilities dependent on the properties of an individual particle) on the basis of macroscopic data in the form of reaction rate coefficients for each surface process. Thus, we use the information contained in macroscopic surface reaction sets to develop DSMC surface chemistry models. The approach to the extraction of DSMC parameters from macroscopic ones will be proposed in this section. The models described below are applicable to any implementation of the DSMC method.

In the procedure of obtaining the microscopic properties from the macroscopic ones, we assume the molecular velocity distribution function in the vicinity of the surface to be equilibrium. This assumption is similar to those made in the total collsion energy model of gas-phase reactions. It also seems to be reasonable for the surface processes because the considered macroscopic models are applicable to the continuum flow regime where the distribution function is close to equilibrium.

We assume that the body surface is presented in the DSMC computations as a set of elements of a known area, which are small as compared to the macroscopic parameter gradient length scale (in the direction parallel to the surface). The approach is different for surface and impact mechanisms; therefore, these processes will be considered separately.

A. Surface mechanisms

Surface mechanisms involve only particles adsorbed on the surface. Let us consider a certain process P that involves adsorbed particles A_S. The surface number density of adsorbed particles n_{A_S} changes according to the equation

$$\frac{dn_{A_S}}{dt} = -K_P n_R, \tag{3}$$

where K_P is the reaction rate coefficient of the process P and n_R is the surface number density of a reagent or reagents (the number of reagents depends on the reaction type under consideration). The frequency of the process ν_P can be determined from this equation. Assuming that the elementary events of the process P follow the Poisson process and hence time t_p until the next elementary event follows the exponential distribution, we can sample this time in the DSMC using the formula

$$t_P = -\frac{\ln(R_n)}{\nu_P}, \tag{4}$$

where R_n is a random number between 0 and 1. Such an approach is used in the null-collision (Refs. 52 and 53) and majorant frequency schemes (Ref. 54) for Monte Carlo simulation of gases and plasmas.

1. Desorption

Let us consider the desorption reaction

$$A_S \rightarrow A + S, \tag{5}$$

where A is an arbitrary particle (an atom or a molecule). Equation (3) characterizing a change in the number density of adsorbed particles n_{A_S} is rewritten for the desorption process as

$$\frac{dn_{A_S}}{dt} = -K_{des} n_{A_S}. \tag{6}$$

Desorption is an individual process, i.e., it does not involve other adsorbed particles. The frequency of desorption per one particle is

$$\nu_{des} = K_{des}. \tag{7}$$

The characteristic time of desorption here is the mean lifetime of a particle A_S on the surface. The time spent by each adsorbed particle on the surface can be sampled using the formula (4) as

$$\tau = -\frac{\ln(R_n)}{K_{des}}. \tag{8}$$

For each particle that appears on the surface in a DSMC calculation after the adsorption process or after the Langmuir-Hinshelwood recombination, the lifetime of this particle on the surface is sampled with the formula (8). After this time, the particle should be desorbed from the surface: desorption is performed at the end of each time step by checking the time spent on the surface for each adsorbed particle. The desorbed particle that moved to the gas phase is assumed to be reflected diffusely from the surface with the distribution corresponding to the wall temperature T_w.

2. Langmuir-Hinshelwood recombination

Let us consider the Langmuir-Hinshelwood recombination

$$A_S + B_S \rightarrow AB + 2S, \tag{9}$$

where A and B are arbitrary atoms (or molecules) and AB is a corresponding molecule.

Equation (3) characterizing a change in the number density of adsorbed particles is rewritten for the Langmuir-Hinshelwood recombination as

$$\frac{dn_{A_S}}{dt} = \frac{dn_{B_S}}{dt} = -K_{LH} n_{A_S} n_{B_S}. \tag{10}$$

The frequency $\tilde{\nu}_{LH}$ of real Langmuir-Hinshelwood reactions, which is the number of reactions per unit time, can be derived from formula (10) in the following way:

$$\tilde{\nu}_{LH} = K_{LH} n_{A_S} n_{B_S} S_e. \tag{11}$$

S_e is the area of the surface element, on which the reaction is considered. Particles on the surface interact with each other independently of their location on the surface.

Applying the standard majorant frequency scheme (Ref. 54) of the DSMC method and assuming equal recombination probabilities for all the pairs of particles adsorbed on the surface element with the area S_e, we can rewrite the previous formula (11) in terms of model particles, i.e., obtain the frequency ν_{LH} of the Langmuir-Hinshelwood reactions between model particles

$$\nu_{LH} = \tilde{\nu}_{LH} \frac{1}{F_N} = K_{LH} N_{A_S} N_{B_S} \frac{{F_N}^2}{{S_e}^2} S_e \frac{1}{F_N} = K_{LH} N_{A_S} N_{B_S} \frac{F_N}{S_e}, \tag{12}$$

where F_N is the number of real particles represented by one model particle in the DSMC method and N_{A_S} and N_{B_S} are the number of model particles of respective species adsorbed on the surface element with the area S_e.

For the special case of the reagents of the same species $A_S + A_S \rightarrow A_2 + 2S$, one obtains

$$\begin{aligned} \nu_{LH} &= \tilde{\nu}_{LH} \frac{1}{F_N} = \frac{1}{2} K_{LH} N_{A_S} (N_{A_S} - 1) \frac{{F_N}^2}{{S_e}^2} S_e \frac{1}{F_N} \\ &= \frac{1}{2} K_{LH} N_{A_S} (N_{A_S} - 1) \frac{F_N}{S_e}. \end{aligned} \tag{13}$$

Applying formula (4), we can sample the time between the consecutive model Langmuir-Hinshelwood recombination events from the exponential distribution

$$t_{LH} = -\frac{\ln(R_n)}{\nu_{LH}}. \tag{14}$$

The frequency should be recalculated and the time between the events should be resampled each time the number of adsorbed molecules of any reacting species is changed. The Langmuir-Hinshelwood reactions that occur on the surface during the DSMC time step Δt are performed at the end of each time step on each surface element consecutively.

The resulting molecule that is the product of the Langmuir-Hinshelwood recombination can stay adsorbed on the surface or move to the gas phase after the reaction. In the first case, the recombination energy E_{recomb} produced in the reaction is completely accommodated by the surface, i.e., contributes to the heat flux to the surface. In the second case, different degrees of recombination energy accommodation can be considered. In the case of complete accommodation, similar to the first case, the entire energy contributes to the heat flux, whereas the molecule that is the reaction product is diffusely reflected with the distribution of all energy modes corresponding to the wall temperature T_w. In the case of incomplete accommodation of recombination energy, some part of this energy contributes to the heat flux, whereas the remaining part should be added to the energy of the reflected molecule that is the reaction product.

The degree of accommodation of the chemical energy on the surface has been addressed in several experimental and theoretical studies (see, e.g., Refs. 55 and 56). It should be noted that the Larsen-Borgnakke approach (Ref. 57) can be naturally used for the energy distribution between the translational, rotational, and vibrational modes in the framework of the present approach. In the present paper, we consider only the case of complete accommodation of recombination energy. As it has been pointed out above, the present DSMC approach is based on a macroscopic finite-rate surface recombination models. As was stated in the recent review (Ref. 58), these macroscopic models do not explicitly address energy accommodation. Normally, the complete energy accommodation is assumed and thus the predicted heating rate for a particular surface chemistry model is maximized. The present paper follows the same logic and one should consider the present DSMC results on the heating rate as the maximum possible value for the particular reaction mechanism. The question of incomplete accommodation of the recombination energy within the proposed DSMC approach should be addressed in the further studies.

B. Impact mechanisms

Impact mechanisms involve particles A_S adsorbed on the surface and gas-phase particles B colliding with this surface. Therefore, the DSMC model parameters of interest have to depend on the individual properties of the particle B that arrives from the gas phase as well as on the surface density n_{A_S} of adsorbed particles A_S. For impact mechanisms, the microscopic probabilities of the processes will be determined in this section.

Let us consider a certain impact process that involves a gas-phase particle B with a velocity V colliding with the surface. The microscopic probability P of the process is assumed to depend on the velocity component V_n normal to the surface. Then the macroscopic probability of this process $\bar{P}$, which depends on the gas temperature T_g, can be expressed as the average of the microscopic probability P in the following manner:

$$\bar{P}\left(T_g\right) = \frac{\int_0^\infty P(V_n) V_n f\left(V_n, T_g\right) dV_n}{\int_0^\infty V_n f\left(V_n, T_g\right) dV_n}. \tag{15}$$

Let us assume that the distribution function $f(V_n, T_g)$ is at equilibrium

$$f\left(V_n, T_g\right) = \left(\frac{m_B}{2\pi k T_g}\right)^{\frac{1}{2}} \exp\left(-\frac{m_B V_n{}^2}{2kT_g}\right). \quad (16)$$

Applying the change of variables

$$\begin{cases} \varepsilon = \frac{m_B}{2kT_g} \\ \varphi = V_n{}^2 \end{cases} \Leftrightarrow \begin{cases} T_g = \frac{m_B}{2k\varepsilon} \\ V_n = \varphi^{\frac{1}{2}} \end{cases}, \quad (17)$$

we can rewrite formula (15) as

$$\frac{1}{\varepsilon}\bar{P}\left(\frac{m_B}{2k\varepsilon}\right) = \int_0^\infty P\left(\varphi^{\frac{1}{2}}\right)\exp(-\varphi\varepsilon)d\varphi = \mathcal{L}\left\{P\left(\varphi^{\frac{1}{2}}\right)\right\}, \quad (18)$$

where $\mathcal{L}$ is the Laplace transform, which converts the function depending on the variable φ to a function depending on the variable ε. Therefore, the microscopic probability P of the process can be expressed through the macroscopic probability $\bar{P}$ using the inverse Laplace transform as follows:

$$P\left(\varphi^{\frac{1}{2}}\right) = \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\bar{P}\left(\frac{m_B}{2k\varepsilon}\right)\right\}. \quad (19)$$

Thus, the next step is to define the macroscopic probability $\bar{P}$ of the process using the known reaction rate constant specified in the initial macroscopic mechanism. Then the microscopic probability P for the DSMC method can be determined analytically with the use of formula (19) and inverse Laplace transform properties.

The variation of the surface number density of adsorbed particles n_{A_S} is described by the equation

$$\frac{dn_{A_S}}{dt} = -Kn_B n_R, \quad (20)$$

where K is the reaction rate coefficient of the process, n_B is the volume number density of particles B colliding with the surface and taking part in the considered surface reaction, and n_R is the surface number density of other reagent if it exists (the number of reagents depends on the reaction type). The macroscopic probability $\bar{P}$ of the process is defined as

$$\bar{P} = \frac{J_B^{reac}}{J_B}, \quad (21)$$

where J_B is the flux of particles B incident onto the surface and J_B^{reac} is the flux of particles B incident onto the surface and taking part in the reaction. The total flux of particles to the surface in equilibrium can be written as (Ref. 2)

$$J_B = \frac{1}{4}\sqrt{\frac{8kT_g}{\pi m_B}} n_B, \quad (22)$$

and the flux of particles to the surface participating in the reaction can be obtained from Eq. (20)

$$J_B^{reac} = -\frac{dn_{A_S}}{dt} = Kn_B n_R. \quad (23)$$

Therefore, substituting Eqs. (22) and (23) into formula (21), we can obtain the required dependence of the macroscopic probability $\bar{P}$ on the gas temperature T_g

$$\bar{P} = \frac{J_B^{reac}}{J_B} = \frac{Kn_B n_R}{\frac{1}{4}\sqrt{\frac{8kT_g}{\pi m_B}} n_B} = \sqrt{\frac{2\pi m_B}{kT_g}} K n_R. \quad (24)$$

Various impact mechanisms are considered below.

1. Adsorption

Let us consider the process of adsorption,

$$A + S \rightarrow A_S. \quad (25)$$

Equation (20) describing the change in the surface number density of adsorbed particles n_{A_S} is rewritten as

$$\frac{dn_{A_S}}{dt} = K_{ads} n_A n_S, \quad (26)$$

where n_S is the surface density of free surface sites.

Using Eq. (24), we obtain the macroscopic probability of the adsorption process

$$\bar{P}_{ads} = \frac{J_A^{reac}}{J_A} = \frac{K_{ads} n_A n_S}{\frac{1}{4}\sqrt{\frac{8kT_g}{\pi m_A}} n_A} = \sqrt{\frac{2\pi m_A}{kT_g}} K_{ads} n_S$$
$$= \sqrt{\frac{2\pi m_A}{kT_g}} K_{ads} N_S \frac{F_N}{S_e}, \quad (27)$$

where N_S is the number of free model surface sites on the surface element with the area S_e, i.e., the number of vacant surface sites on which the model particle can be adsorbed.

2. Eley-Rideal recombination

Let us consider the Eley-Rideal recombination

$$A_S + B \rightarrow AB + S. \quad (28)$$

Equation (20) is rewritten as

$$\frac{dn_{A_S}}{dt} = -K_{ER} n_B n_{A_S}. \quad (29)$$

Therefore, Eq. (24) is transformed to

$$\bar{P}_{ER} = \sqrt{\frac{2\pi m_B}{kT_g}} K_{ER} n_{A_S} = \sqrt{\frac{2\pi m_B}{kT_g}} K_{ER} N_{A_S} \frac{F_N}{S_e}. \quad (30)$$

3. Dissociative adsorption

Let us consider the process of dissociative adsorption

$$AB + S \rightarrow A_S + B. \quad (31)$$

Equation (20) is rewritten as

$$\frac{dn_{A_S}}{dt} = K_{disads} n_{AB} n_S. \quad (32)$$

Therefore, Eq. (24) is transformed to

$$\bar{P}_{disads} = \sqrt{\frac{2\pi m_{AB}}{kT_g}} K_{disads} n_S = \sqrt{\frac{2\pi m_{AB}}{kT_g}} K_{disads} N_S \frac{F_N}{S_e}. \quad (33)$$

Thus, the macroscopic probability of the process for all impact mechanisms has the form

$$\bar{P} = \sqrt{\frac{2\pi m}{kT_g}} K \Lambda \quad (34)$$

where Λ is a parameter, which contains information about surface occupation, i.e., the number of free surface sites or the number of adsorbed particles (Λ does not depend on the gas temperature T_g) and K is the reaction rate constant of the process under consideration, which depends on the temperature T; see Eq. (1). We can treat this temperature as the gas temperature T_g or surface temperature T_w. Indeed, under continuum flow conditions, for which the reaction rate constants are typically obtained and where they are applied, the gas temperature can be considered to be equal to the surface temperature. However, in the procedure described above, T_g is the variable of integration, while T_w is constant; therefore, the result of the procedure application depends on our treatment of the temperature.

Let us consider two cases: $T = T_w$ and $T = T_g$. In the first case, the reaction rate constant K is independent of the gas temperature T_g, i.e., of the variable ε defined by formula (17). It means that K is a constant in the inverse Laplace transform (19). In the second case, the opposite situation is observed: K is not a constant in the inverse Laplace transform.

Thus, if we use $T = T_w$ in expression (1) for the reaction rate constant $K(T)$ in Eq. (19), the microscopic probability $P(V_n)$ is determined as

$$\begin{aligned} P(V_n) &= P\left(\varphi^{\frac{1}{2}}\right) = \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\bar{P}\left(\frac{m}{2k\varepsilon}\right)\right\} \\ &= \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\sqrt{4\pi\varepsilon}K(T_w)\Lambda\right\} = 2\sqrt{\pi}K(T_w)\Lambda\mathcal{L}^{-1}\left\{\varepsilon^{-\frac{1}{2}}\right\} \\ &= 2\sqrt{\pi}K(T_w)\Lambda\frac{\varphi^{-\frac{1}{2}}}{\Gamma\left(\frac{1}{2}\right)}\mathcal{H}(\varphi) \\ &= 2\sqrt{\pi}K(T_w)\Lambda\frac{1}{\sqrt{\pi}V_n}\mathcal{H}\left(V_n^2\right) = 2K(T_w)\Lambda\frac{1}{V_n}, \end{aligned} \tag{35}$$

where Γ is the gamma function and $\mathcal{H}$ is the Heaviside step function. It is seen from Eq. (35) that the individual probability of each impact process $P(V_n)$ is inversely proportional to the normal to the surface component of the gas-phase particle velocity and turns to infinity at the zero point [Fig. 2(a)].

However, if we use $T = T_g$ in expression (1) for the reaction rate constant $K(T)$ in Eq. (19), the microscopic probability $P(V_n)$ is determined in a different way

$$\begin{aligned} P(V_n) &= P\left(\varphi^{\frac{1}{2}}\right) = \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\bar{P}\left(\frac{m}{2k\varepsilon}\right)\right\} = \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\sqrt{4\pi\varepsilon}K\left(\frac{m}{2k\varepsilon}\right)\Lambda\right\} = \mathcal{L}^{-1}\left\{2\sqrt{\pi}\varepsilon^{-\frac{1}{2}}A\left(\frac{m}{2k\varepsilon}\right)^b\exp\left(-\frac{2E_a}{m}\varepsilon\right)\Lambda\right\} \\ &= 2\sqrt{\pi}\left(\frac{m}{2k}\right)^b A\Lambda\mathcal{L}^{-1}\left\{\varepsilon^{-\frac{1}{2}-b}\right\}\left(\varphi-\frac{2E_a}{m}\right)\mathcal{H}\left(\varphi-\frac{2E_a}{m}\right) = 2\sqrt{\pi}\left(\frac{m}{2k}\right)^b A\Lambda\frac{\left(\varphi-\frac{2E_a}{m}\right)^{b-\frac{1}{2}}}{\Gamma\left(b+\frac{1}{2}\right)}\mathcal{H}^2\left(\varphi-\frac{2E_a}{m}\right) \\ &= \frac{2\sqrt{\pi}}{\Gamma\left(b+\frac{1}{2}\right)}\left(\frac{m}{2k}\right)^b A\Lambda\left(V_n^2-\frac{2E_a}{m}\right)^{b-\frac{1}{2}}\mathcal{H}^2\left(V_n^2-\frac{2E_a}{m}\right) = \begin{cases} \frac{\sqrt{2\pi m}}{\Gamma\left(b+\frac{1}{2}\right)k^b}A\Lambda\left(\frac{mV_n^2}{2}-E_a\right)^{b-\frac{1}{2}}, & if\ \frac{mV_n^2}{2} > E_a \\ 0, & if\ \frac{mV_n^2}{2} \le E_a \end{cases}. \end{aligned} \tag{36}$$

Note that the derivation of Eq. (36) is correct if $E_a > 0$. Thus, it is seen from Eq. (36) that the microscopic probability of each impact process $P(V_n)$ has a different character of the dependence on the gas-phase particle normal velocity as compared to the previous case [see Figs. 2(b)–2(d)]: the probability is not equal to zero only if the kinetic energy of the gas-phase particle along the normal to the surface exceeds the energy E_a; depending on the value of b, the following options are possible: (i) the probability has a threshold and turns to infinity ($b < 1/2$), (ii) it is a step function ($b = 1/2$), or (iii) the probability increases ($b > 1/2$).

It should be noted that the analytical formulas (35) and (36) may yield probability values greater than unity, for example, near $V_n = 0$ in the case illustrated in Fig. 2(a), near $V_n = \sqrt{\frac{2E_a}{m}}$ in Fig. 2(b), and near $V_n \to \infty$ in Fig. 2(d). It is well known that a similar problem is inherent in phenomenological models of gas-phase chemical reactions (see, e.g., Ref. 59). Applying this probability formula directly in

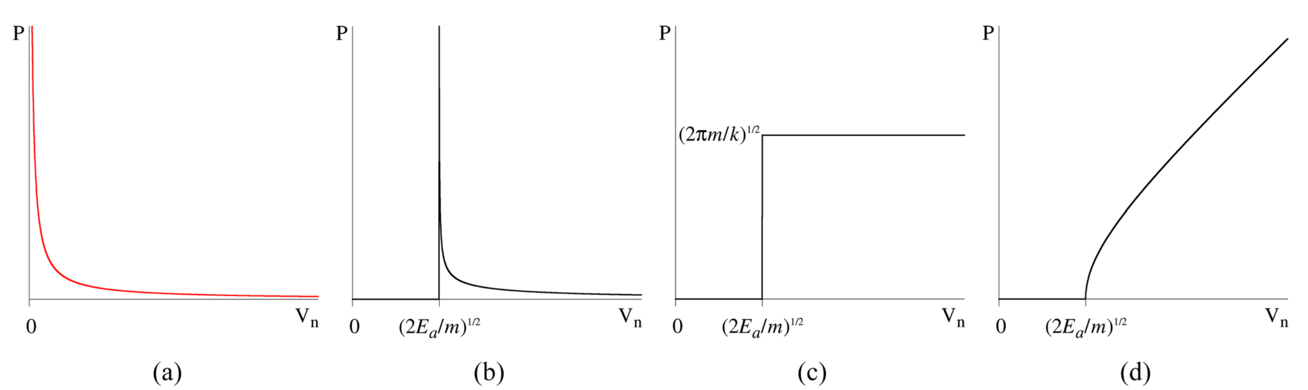


FIG. 2. Microscopic probability of impact processes $P(V_n)$: (a) formula (35); (b) formula (36), $b < 1/2$; (c) formula (36), $b = 1/2$; (d) formula (36), $b > 1/2$.

computations, where its values are compared with the values of a pseudo-random variable uniformly distributed over the segment [0,1], one *de facto* uses a "truncated" probability function at the level of unity. This means that the probability in those regions where the analytical formulas predict its values greater than unity is assigned the unit value there. As a result, the number of impact processes in simulations can be smaller than the number of these processes that could be expected from the initial reaction rates of these processes. The size of error associated with probability values greater than unity depends on the chosen shape of the functional dependence of the probability: Eq. (35) [see Fig. 2(a)] or Eq. (36) [see Figs. 2(b)–2(d)].

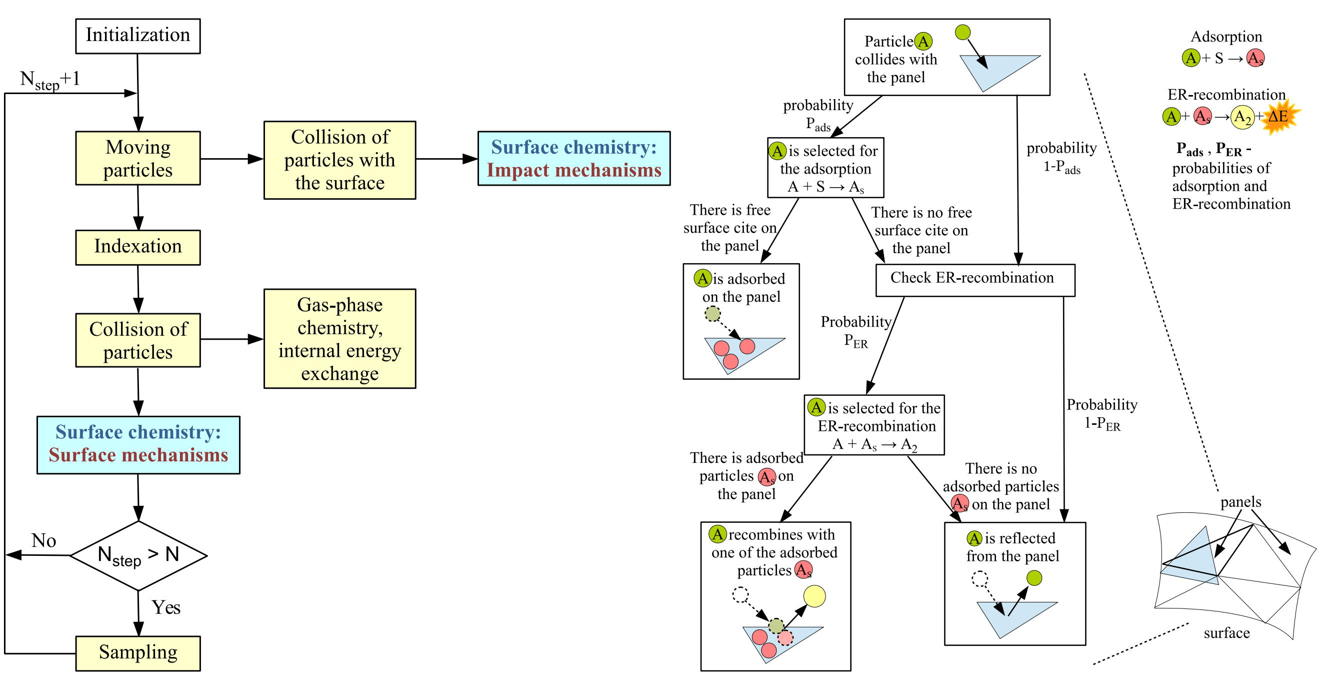
The analysis shows that this error in the case of a threshold determined by Eq. (36) is appreciably higher than that in the case of Eq. (35), which does not contain a threshold. In particular, for the model that describes surface reactions on reaction cured glass (RCG) (see Sec. IV A), the reaction rates predicted by the threshold-type equation for the probability are significantly lower than the initial value, even if each particle with a velocity greater than the threshold value reacted (i.e., when the reaction probability is a stepwise function equal to unity for the particle velocity exceeding the threshold value). It should be also noted that, for gas temperatures near the wall typical for a continuum flow about a reentry vehicle (below 2000 K), this threshold is several times higher than the mean thermal velocity of molecules, i.e., only high-energy particles are involved into the Eley-Rideal recombination, which cannot be clearly explained on the basis of the physics of the process in our opinion. Based on the considerations discussed above, we chose formula (35) for further calculations.

In the case of the Eley-Rideal recombination, as in the case of Langmuir-Hinshelwood recombination, the resulting molecule can stay adsorbed on the surface or leave the surface after the Eley-Rideal reaction. The gas-phase product of recombination is reflected from the surface diffusely with the distribution corresponding to the wall temperature T_w. It is also possible to consider the case on incomplete accommodation of recombination energy E_{recomb} similarly to the Langmuir-Hinshelwood case; in the present paper, however, only the case of complete accommodation of recombination energy is analyzed. Hence, as it was stated in Sec. III C, the present DSMC results provide the maximum value of the heating rate for a particular finite-rate reaction mechanism, and the effect of incomplete recombination energy accommodation should be addressed in the future.

In the case of dissociative adsorption, the gas phase product of the reaction, if it exists, leaves the surface diffusely with the distribution corresponding to the wall temperature T_w. The energy required to break the bonds of the incident molecule in the course of surface dissociation is taken into account in calculating the heat flux (it makes a negative contribution to the thermal energy flux to the surface).

C. Implementation in the DSMC method

Let us consider a body surface approximated by a geometrical model composed of flat elements, e.g., triangles in the 3D case or line segments in the 2D case. In the SMILE++ software system which is used in the present study such flat surface elements are referred as "panels." All surface processes are modeled at each surface element separately. Figure 3(a) illustrates the overall DSMC algorithm including surface processes: the impact mechanisms are modeled during each time step at each event of the collision of a gas-phase particle with the surface, and the surface mechanisms are modeled



(a) (b)

FIG. 3. (a) DSMC scheme including surface processes, (b) scheme of impact mechanisms.

collectively at the end of each time step Δt on each surface element.

The surface mechanisms (Langmuir-Hinshelwood recombination and desorption) are performed in the following iterative process:

$$
\begin{aligned}
& t = \Delta t \\
& while\ t > 0\ do: \\
& \quad \left[\begin{array}{l} LH - recombination \\ t = t - t_{LH} \end{array} \right. \\
& for\ each\ A_S \\
& \quad if\ t_{panel} > t_{des}\ do: \\
& \quad \quad Desorption\ of\ A_S
\end{aligned} \tag{37}
$$

Here t_{LH} is the time of one Langmuir-Hinshelwood recombination calculated by formula (14), t_{panel} is the time spent by the particle on the surface, and t_{des} is the lifetime of the particle on the surface calculated by formula (8).

The algorithm for the impact mechanisms (adsorption and Eley-Rideal recombination) is shown in Fig. 3(b). Each impact process is performed with a probability $P(V_n)$ depending on the gas-phase particle energy and obtained in the Sec. III B. In the case of adsorption, as well as dissociative adsorption, the gas-phase particle can be captured by the surface element if there is a free surface site on it; adsorbed particles are deleted from the gas phase and stored on the surface. In the case of dissociative adsorption, the non-adsorbed product of the reaction, if it exists, is reflected from the surface (it is not shown in the figure). In the case of the Eley-Rideal recombination, the partner for the recombination for the gas-phase particle colliding with some surface element is chosen randomly from particles adsorbed on the same surface element, and the resulting molecule is reflected from the surface.

IV. DSMC SURFACE CHEMISTRY MODELS

The proposed approach for obtaining DSMC surface chemistry models was applied to two macroscopic surface reaction sets: the well-known mechanism of Warnatz *et al.* (Ref. 33) of chemical processes on a reaction cured glass (RCG)-surface in air and the recent mechanism of Kovalev *et al.* (Ref. 40) of chemical processes on an α-Al_2O_3-surface in air. In these macroscopic mechanisms, a five-species mixture of gases (N, O, N_2, O_2, NO) is considered, and a detailed mechanism of heterogeneous catalytic reactions in dissociated air is proposed. The reaction rate coefficients are specified for each considered surface process. These mechanisms differ not only in the material of the surface to which they are applied, but also in the method of obtaining the reaction rate coefficients. In the first surface reaction set, the reaction rate constants were obtained from experimental data, whereas the reaction rate constants in the second set were obtained on the basis of quantum-mechanical calculations. The probabilities of the processes can be also obtained by the method of Ref. 40; however, they were not presented in the paper.

Note that the present approach can be also applied to other macroscopic surface reaction sets in a similar manner. A sufficient condition for applying the approach to a surface reaction set can be formulated as follows: the reaction mechanism has to be presented as a set of elementary processes considered in the Sec. III, and the reaction rate constants of these processes have to be presented in the Arrhenius form.

A. Catalysis on an RCG-surface, mechanism of Warnatz *et al.*

Let us first consider the surface reaction set of Warnatz *et al.* (Ref. 33) of chemical processes on an RCG-surface in an air mixture. RCG (reaction cured glass, consists of SiO_2 by 94%) is a low catalytic material frequently used for production of the reusable spacecraft thermal protection system (TPS), for example, in the space shuttle surface coating. The following surface processes are taken into account (see Table II): adsorption of N and O atoms on the surface, formation of N_2, and O_2 molecules in the Eley-Rideal and Langmuir-Hinshelwood recombination (formation of NO is neglected), and desorption of N and O atoms, and also N_2 and O_2 molecules that remain adsorbed after the Langmuir-Hinshelwood recombination from the surface. The reaction rate constants for desorption, Eley-Rideal, and Langmuir-Hinshelwood reactions are defined in the Arrhenius form (1), $b = 0$; for adsorption reactions, a constant sticking coefficient S_0 is specified. The parameters for the reaction rate constants and sticking coefficients are listed in Table II.

Using the approach proposed above, we obtain the characteristic times of the surface processes and the individual probabilities of the impact processes in DSMC computations from their reaction rate coefficients.

For *desorption* processes listed in Table II, we calculate the lifetime of an adsorbed particle on the surface, after which the particle should be desorbed to the gas phase, according to Eq. (8)

$$\tau = -\frac{\ln(R_n)}{K_{des}}. \tag{38}$$

TABLE II. Parameters of reactions in macroscopic surface reaction set of Ref. 33. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

Reaction	S_0	A	E_a
Adsorption			
$N + S \rightarrow N_S$	0.1		
$O + S \rightarrow O_S$	0.1		
Desorption		(1/s)	(J)
$N_S \rightarrow N + S$		7.3×10^{11}	3.6×10^{-19}
$O_S \rightarrow O + S$		5×10^{11}	3.3×10^{-19}
$N_{2S} \rightarrow N_2 + S$		1×10^{12}	0.17×10^{-19}
$O_{2S} \rightarrow O_2 + S$		1×10^{12}	0.17×10^{-19}
Eley-Rideal		(m^3/s)	(J)
$N + N_S \rightarrow N_2 + S$		0.4×10^{-16}	0.66×10^{-19}
$O + O_S \rightarrow O_2 + S$		1×10^{-16}	1×10^{-19}
Langmuir-Hinshelwood		(m^2/s)	(J)
$N_S + N_S \rightarrow N_{2S} + S$		1.16×10^{-10}	2.16×10^{-19}
$O_S + O_S \rightarrow O_{2S} + S$		33×10^{-10}	2.66×10^{-19}

For each *Langmuir-Hinshelwood recombination reaction* listed in Table II, the time between two consecutive reactions is calculated according to Eqs. (14) and (13) by the formula

$$t_{LH} = -\frac{\ln(R_n)}{\nu_{LH}}, \quad \text{where } \nu_{LH} = \frac{1}{2}K_{LH}N_{A_S}(N_{A_S}-1)\frac{F_N}{S_e}. \tag{39}$$

In the case of *adsorption,* a constant sticking coefficient S_0 is specified. It means that formula (27) in the model under consideration has a simpler form

$$\bar{P}_{ads} = S_0 = 0.1. \tag{40}$$

Thus, using formula (19) and inverse Laplace transform properties, we can find the microscopic probability P_{ads} of adsorption for the DSMC method

$$P_{ads}(V_n) = P_{ads}\left(\varphi^{\frac{1}{2}}\right) = \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\bar{P}_{ads}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}S_0\right\}$$
$$= S_0\mathcal{L}^{-1}\left\{\frac{1}{\varepsilon}\right\} = S_0\mathcal{H}(\varphi) = S_0, \tag{41}$$

i.e., the microscopic probability for each gas-phase particle colliding with the surface to be adsorbed is equal to the same given constant sticking coefficient S_0.

For the *Eley-Rideal recombination reaction,* we can calculate the microscopic probability combining Eqs. (35) and (30). The resulting individual probability of the reaction is

$$P_{ER}(V_n) = 2K_{ER}(T_w)N_{A_S}\frac{F_N}{S_e}\frac{1}{V_n}. \tag{42}$$

B. Catalysis on an α-Al_2O_3-surface, mechanism of Kovalev *et al.*

Let us consider the surface reaction set of Kovalev *et al.* (Ref. 40) of chemical processes on an α-Al_2O_3-surface in an air mixture. Adsorption of nitrogen is neglected in this mechanism; therefore, only oxygen recombination, as well as adsorption and desorption, are taken into account. The reaction rate constants for all processes are defined in the Arrhenius form (1). The full mechanism of surface processes, as well as parameters for their reaction rate coefficients are presented in Table III.

Applying the proposed approach to this macroscopic surface reaction set, we obtain the model of surface chemistry on an α-Al_2O_3-surface in an air mixture for the DSMC method.

TABLE III. Parameters of reactions in the macroscopic surface reaction set of Ref. 40. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

Reaction	A	b	E_a
Adsorption	(m^3/s)		(J)
$O + S \rightarrow O_S$	6.3×10^{21}	1	-6.2×10^{-21}
Desorption	(1/s)		(J)
$O_S \rightarrow O + S$	4.8×10^{11}	0	2.7×10^{-19}
Eley-Rideal	(m^3/s)		(J)
$O + O_S \rightarrow O_2 + S$	5.2×10^{-18}	1/2	0.69×10^{-19}
Langmuir-Hinshelwood	(m^2/s)		(J)
$O_S + O_S \rightarrow O_2 + 2S$	2.7×10^{-7}	0	4.0×10^{-19}

The characteristic times of the surface processes and the individual probabilities of the impact processes for the DSMC computations are calculated almost in the same manner as in the previous case and are presented below. The lifetime of the adsorbed particle on the surface for the *desorption* process is

$$\tau = -\frac{\ln(R_n)}{K_{des}}, \tag{43}$$

and the time between two consecutive *Langmuir-Hinshelwood recombination* reactions is

$$t_{LH} = -\frac{\ln(R_n)}{\nu_{LH}}, \quad \text{where } \nu_{LH} = \frac{1}{2}K_{LH}N_{A_S}(N_{A_S}-1)\frac{F_N}{S_e}. \tag{44}$$

The microscopic probability of *adsorption* for the α-Al_2O_3-model differs from the previous case and is calculated by combining Eqs. (35) and (27) as

$$P_{ads}(V_n) = 2K_{ads}(T_w)N_S\frac{F_N}{S_e}\frac{1}{V_n}, \tag{45}$$

and finally, the microscopic probability of *Eley-Rideal recombination* is

$$P_{ER}(V_n) = 2K_{ER}(T_w)N_{A_S}\frac{F_N}{S_e}\frac{1}{V_n}. \tag{46}$$

V. RESULTS OF DSMC COMPUTATIONS AND DISCUSSION

The molecular models of catalytic processes on different surfaces proposed above are similar in terms of the numerical algorithm: the only difference between them is the values of the reaction rate coefficients. Therefore, these models can be formulated in a general form. This general model was implemented in SMILE++, a robust DSMC software system developed at ITAM, which provides a full cycle of numerical studies on multiprocessor computers, from definition of a three-dimensional geometric model of the vehicle to the visualization of computational results. The verification of the implemented algorithm is presented below. Then the results of application of the model to assessing surface catalycity effects on aerothermodynamics of bodies under reentry flow conditions are discussed.

A. Verification

The following series of computations were performed to compare the analytical values of the reaction rate constants with those numerically calculated by SMILE++ in the conditions of thermally equilibrium gas with temperature equal to the surface one (both gas and surfaces gas temperature was denoted here by T). Due to the fact that the DSMC model parameters are obtained under these thermal equilibrium conditions from the known macroscopic rates, one should expect almost exact agreement of the DSMC results with the theory if the DSMC model was implemented correctly. The verification was performed for the case of oxygen processes on an RCG surface (in terms of the numerical algorithm, the cases of all gases and surface materials are similar). The DSMC calculations were performed for a 2D square computational domain with three specular boundaries (left, top, and bottom) and one diffuse wall (placed at the right of the domain) on which the considered RCG-surface processes (see Table II) were modeled. The computational domain was filled with atomic oxygen. The gas pressure was 4 Pa, and the temperature T was varied from the minimum (700 K) to maximum (2000 K) value in steps of 100 K. A separate DSMC computation was carried out for each value of the temperature.

Figure 4(a) shows the desorption rate constant for atomic oxygen. In this DSMC calculation, only the desorption process was modeled and the other three surface processes were neglected. At the start of each computation, a set of adsorbed particles was generated on the surface. During the computation, desorption of particles from the surface was performed after each time step, and the lifetime of each particle on the surface was calculated. These values were sampled to calculate the mean lifetime $\bar{\tau}$ of particles on the surface and the frequency of desorption ν_{des}, which is equal to the desorption rate constant K_{des} [Eq. (7)],

$$K_{des} = \nu_{des} = \frac{1}{\bar{\tau}}. \qquad (47)$$

Note that no other surface processes were modeled in computations, which means that new particles did not appear on the surface by adsorption and did not disappear from the surface in the Eley-Rideal or Langmuir-Hinshelwood recombination reactions.

The Langmuir-Hinshelwood recombination rate constant for the $O_S + O_S \rightarrow O_{2_S} + S$ reaction on an RCG-surface is presented in Fig. 4(b). In the DSMC calculations, only the Langmuir-Hinshelwood recombination was considered: the impact processes and desorption were not modeled. At the start of each computation, a set of adsorbed particles was generated on the surface (N_{O_S} model particles). During the computation, the number of accepted Langmuir-Hinshelwood recombination events on the wall was sampled after each time step, but the reactions were not modeled, i.e., the adsorbed particles generated in the beginning of the calculation were not removed from the surface and did not appear on the surface either because the adsorption process was neglected and, hence, N_{O_S} remained constant. At the end of each computation, the mean number $\bar{N}_{LH}$ of accepted recombination events per time step Δt was

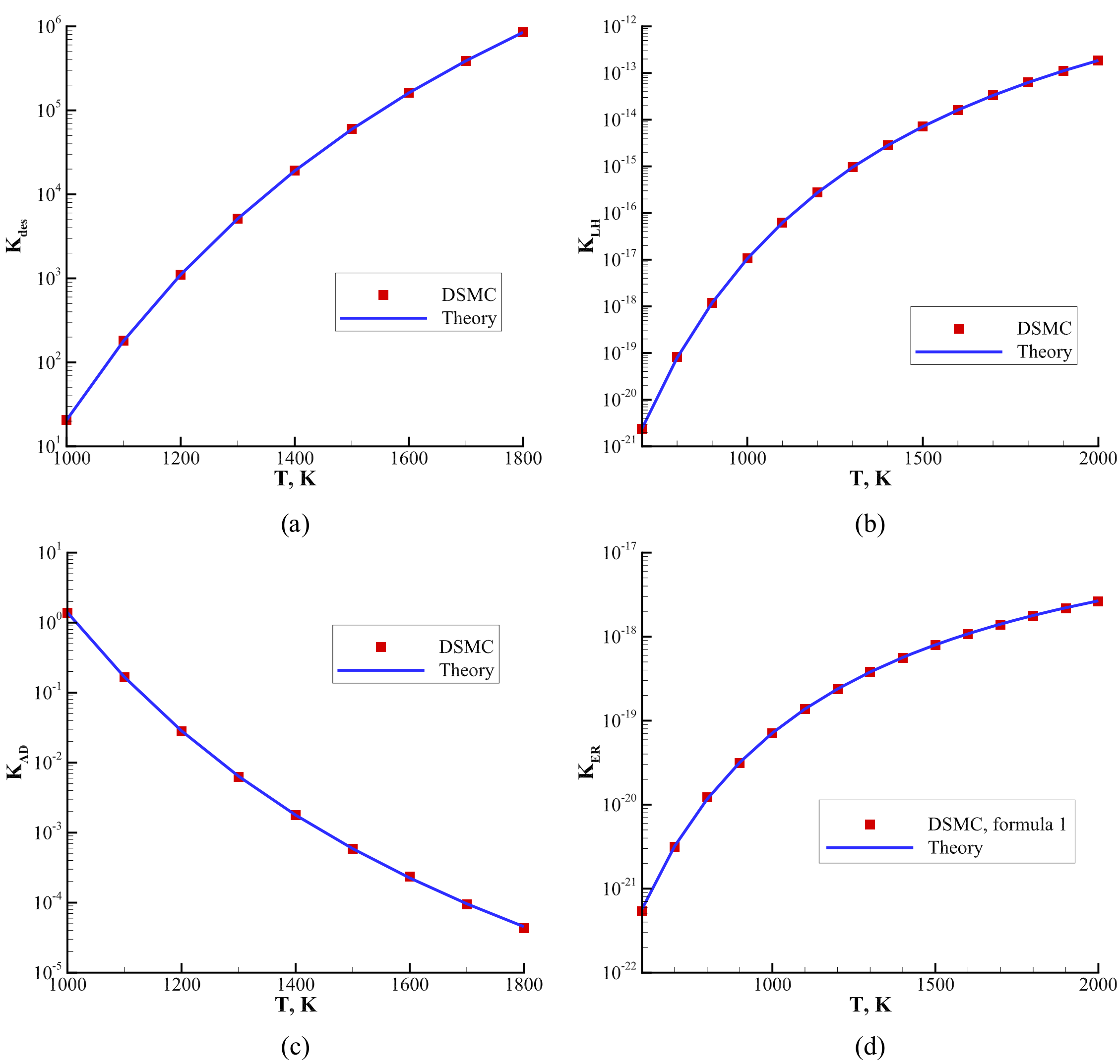


FIG. 4. Comparison of the calculated and analytical reaction rate constants for oxygen processes on the RCG-surface: (a) desorption rate coefficient; (b) Langmuir-Hinshelwood recombination rate coefficient; (c) adsorption-desorption equilibrium constant; (d) Eley-Rideal recombination rate coefficient. Reproduced with permission from Molchanova (Shumakova) *et al.*, AIP Conf. Proc. **1628**, 131 (2014). Copyright 2016 AIP Publishing LLC.

calculated, and then the frequency of recombination ν_{LH} and the reaction rate constant K_{LH} were obtained using Eq. (13):

$$K_{LH} = \nu_{LH}\frac{2}{N_{O_S}(N_{O_S}-1)}\frac{S_{wall}}{F_N} = \frac{\bar{N}_{LH}}{\Delta t}\frac{2}{N_{O_S}(N_{O_S}-1)}\frac{S_{wall}}{F_N} \tag{48}$$

(S_{wall} is the surface area).

The adsorption-desorption equilibrium constant K_{AD} for atomic oxygen is presented in Fig. 4(c). In the DSMC calculations, desorption and adsorption without any recombination were modeled. In that case, the surface was empty at the beginning of each DSMC computation, and the mean number of model particles adsorbed on the surface $\bar{N}_{O_S}$ was calculated during the computation after reaching the steady state. The analytical value of the equilibrium constant K_{AD} is defined by formula (50) derived from the chemical kinetics equation (49) describing the equilibrium adsorption-desorption process $O + S \rightleftarrows O_S$ with the use of the Arrhenius form of the desorption constant

$$\frac{dn_{O_S}}{dt} = S_0\frac{1}{4}\sqrt{\frac{8kT}{\pi m_O}}n_O - K_{des}n_{O_S} = K_{ads}n_O - K_{des}n_{O_S} = 0, \tag{49}$$

$$K_{AD} = \frac{K_{ads}}{K_{des}} = \frac{S_0\frac{1}{4}\sqrt{\frac{8kT}{\pi m_O}}}{K_{des}} = \frac{S_0}{A_{des}}\sqrt{\frac{k}{2\pi m_O}}T^{\frac{1}{2}}\exp\left(\frac{E_{des}}{kT}\right). \tag{50}$$

The numerical value of the equilibrium constant K_{AD} in the DSMC computations is calculated by (51), which follows from Eq. (49), with the calculated mean number of adsorbed particles being $\bar{N}_{O_S}$

$$K_{AD} = \frac{n_{O_S}}{n_O} = \frac{\bar{N}_{O_S}}{n_O}\frac{F_N m_O}{S_{wall}}. \tag{51}$$

Figure 4(d) shows the Eley-Rideal recombination rate constant for the $O + O_S \rightarrow O_2 + S$ reaction on an RCG-surface. In the DSMC calculations, all the surface processes except for the Eley-Rideal recombination were neglected. As in the first two series of computations, N_{O_S} adsorbed model particles were generated on the surface at the beginning of each computation. During the computation, the number of all collisions with the surface N_{coll} and the number of accepted Eley-Rideal recombination events N_{ER} were sampled, but the reactions were not performed, i.e., adsorbed particles were not removed from the surface and gas-phase particles were reflected from the surface without any chemical changes according to the diffuse reflection law (it means that N_{O_S} remained constant). Using these data, the macroscopic probability $\bar{P}_{ER}$ of the Eley-Rideal recombination was calculated by Eq. (21), i.e., N_{ER} divided by N_{coll}, and the Eley-Rideal recombination rate constant was obtained by using Eq. (30)

$$K_{ER} = \sqrt{\frac{kT}{2\pi m_O}}\bar{P}_{ER}\frac{1}{N_{O_S}}\frac{S_{wall}}{F_N} = \sqrt{\frac{kT}{2\pi m_O}}\frac{N_{ER}}{N_{coll}}\frac{1}{N_{O_S}}\frac{S_{wall}}{F_N}. \tag{52}$$

The results of verification are presented in Fig. 4. As it was mentioned above, in this section the proposed model was constructed in such a way that it must capture the reaction rate constants under thermal equilibrium conditions. It is seen that the DSMC method predicts the constants fairly well for the entire considered temperature range for all surface processes.

For more comprehensive verification of code procedures that involve surface chemistry, it is necessary to consider as the test case with simultaneous simulations of all processes considered above. The test case conditions were identical to test conditions for the previous situations. Oxygen atoms incident onto the surface could be adsorbed and participate in the Eley-Rideal recombination reactions with other adsorbed atoms of oxygen. In addition to participation in the Eley-Rideal recombination, the adsorbed atoms could be desorbed and participate in the Langmuir-Hinshelwood recombination reaction. Oxygen molecules formed in the recombination reaction were removed from the computation as soon as they left the surface. As a result, the recombination probabilities were calculated by the formulas

$$\gamma_{LH} = \frac{N_{LH}}{N_{coll}},\ \gamma = \frac{N_{LH}+N_{ER}}{N_{coll}}, \tag{53}$$

where γ denotes the total probability of recombination, γ_{LH} is the probability of only Langmuir-Hinshelwood recombination, N_{coll} is the number of all collisions with the surface, and N_{ER} and N_{LH} are the numbers of all particles accepted for Eley-Rideal and Langmuir-Hinshelwood recombination reactions, respectively, which were sampled during the computation. Verification was performed through comparisons with analytical values of these variables for different values of temperature.

To obtain the analytical values of the total and Langmuir-Hinshelwood probabilities of recombination, the following differential chemical kinetics equation (54) defining the surface number density of adsorbed atoms n_{O_S} in the macroscopic surface reactions mechanism of Warnatz *et al.* was solved analytically

$$\frac{dn_{O_S}}{dt} = S_0\frac{1}{4}\sqrt{\frac{8kT}{\pi m_O}}n_O - K_{des}n_{O_S} - K_{LH}n_{O_S}n_{O_S} - K_{ER}n_{O_S}n_O. \tag{54}$$

Using the definitions of γ_{LH} and γ_{ER} and applying Eq. (2), we transform the third and fourth terms in the right-hand side of this equation

$$K_{LH}n_{O_S}n_{O_S} = J_O^{reac,LH} = \gamma_{LH}\ J_O = \gamma_{LH}\frac{1}{4}\sqrt{\frac{8kT}{\pi m_O}}n_O, \tag{55}$$

$$K_{ER}n_{O_S}n_O = J_O^{reac,R} = \gamma_{ER}\ J_O = \gamma_{ER}\frac{1}{4}\sqrt{\frac{8kT}{\pi m_O}}n_O. \tag{56}$$

Then, substituting the resultant expressions (55) and (56) to the initial differential equation (54), we obtain the following linear heterogeneous differential equation with constant coefficients, which defines the surface population $n_{O_S}(t)$

$$\frac{dn_{O_S}}{dt} + K_{des}n_{O_S} = \sqrt{\frac{kT}{2\pi m_O}}n_O(S_0 - \gamma_{LH} - \gamma_{ER}). \tag{57}$$

The solution of this equation is the function

$$n_{\mathrm{O_S}} = C\exp(-K_{des}t) + \sqrt{\frac{kT}{2\pi m_{\mathrm{O}}}}n_{\mathrm{O}}\frac{(S_0 - \gamma_{LH} - \gamma_{ER})}{K_{des}}, \quad (58)$$

where C is an arbitrary constant. Passing in this solution to the limit as $t \to \infty$, we obtain the steady-state value of the surface population $n_{\mathrm{O_S}}^{eq}$

$$n_{\mathrm{O_S}}^{eq} = \lim_{t\to\infty} n_{\mathrm{O_S}} = \sqrt{\frac{kT}{2\pi m_{\mathrm{O}}}}n_{\mathrm{O}}\frac{(S_0 - \gamma_{LH} - \gamma_{ER})}{K_{des}}. \quad (59)$$

Using Eqs. (55) and (56), we can also obtain the formulas relating the steady-state values of the recombination probabilities γ_{LH} and γ_{ER} to the steady-state value of the surface population $n_{\mathrm{O_S}}^{eq}$

$$\gamma_{LH} = K_{LH}n_{\mathrm{O_S}}^{eq\,2}\bigg/\left(\sqrt{\frac{kT}{2\pi m_{\mathrm{O}}}}n_{\mathrm{O}}\right) = K_{LH}\sqrt{\frac{2\pi m_{\mathrm{O}}}{kT}}\frac{n_{\mathrm{O_S}}^{eq\,2}}{n_{\mathrm{O}}}, \quad (60)$$

$$\gamma_{ER} = K_{ER}n_{\mathrm{O_S}}^{eq}n_{\mathrm{O}}\bigg/\left(\sqrt{\frac{kT}{2\pi m_{\mathrm{O}}}}n_{\mathrm{O}}\right) = K_{ER}\sqrt{\frac{2\pi m_{\mathrm{O}}}{kT}}n_{\mathrm{O_S}}^{eq}. \quad (61)$$

Substituting Eqs. (60) and (61) into Eq. (59), we obtain the quadratic equation for $n_{\mathrm{O_S}}^{eq}$

$$K_{LH}n_{\mathrm{O_S}}^{eq\,2} + (K_{des} + K_{ER}n_{\mathrm{O}})n_{\mathrm{O_S}}^{eq} - S_0\sqrt{\frac{kT}{2\pi m_{\mathrm{O}}}}n_{\mathrm{O}} = 0, \quad (62)$$

whose positive solution is

$$n_{\mathrm{O_S}}^{eq} = \frac{-(K_{des} + K_{ER}n_{\mathrm{O}}) + \sqrt{(K_{des} + K_{ER}n_{\mathrm{O}})^2 + 4K_{LH}S_0\sqrt{\frac{kT}{2\pi m_{\mathrm{O}}}}n_{\mathrm{O}}}}{2K_{LH}}. \quad (63)$$

Thus, we can obtain the analytical steady-state values of the recombination probabilities γ_{LH}, γ_{ER}, and $\gamma = \gamma_{LH} + \gamma_{ER}$ by formulas (60) and (61), where $n_{\mathrm{O_S}}^{eq}$ is determined by Eq. (63).

The probabilities of the total and Langmuir-Hinshelwood recombination γ and γ_{LH} obtained in the DSMC computations are compared with their analytical values in Fig. 5 for different values of temperature (the gas temperature coincides with the surface temperature). The calculated and analytical values are denoted by the red points and blue lines, respectively. The values of the calculated probabilities are in good agreement with the analytical results. Based on the data reported in this section, we can conclude that the general DSMC surface chemistry model is implemented into the SMILE++ code correctly.

B. Comparison with *ab initio* calculation data

The results obtained by using the developed DSMC model of heterogeneous processes were compared with molecular

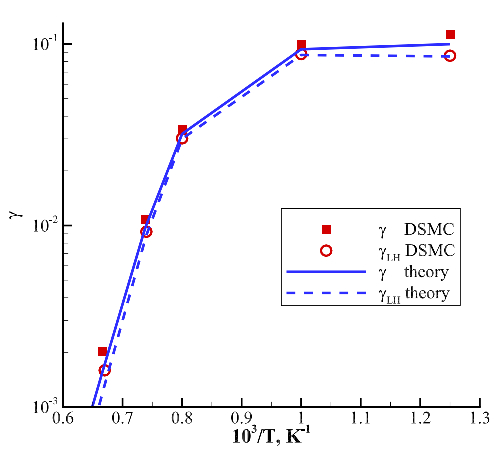


FIG. 5. Comparison of the calculated probability of recombination with the analytical value.

dynamic (MD) calculations of Rutigliano *et al.* (Ref. 60). In Ref. 60, the calculations of the Eley-Rideal recombination of oxygen on a β-cristobalite surface were performed. The surface temperature was assumed to be $T_w = 1000$ K. The probability of an oxygen atom impinging normally onto the surface to be involved in the Eley-Rideal recombination on a small unit cell of the surface crystal lattice was given as a function of kinetic energy of the atom. However, the data on the reaction rate constant as a function of temperature were not provided, and we cannot apply the presented model to this case directly. Nevertheless, the data of Ref. 60 can be used with some constraints to validate the proposed approach for the Eley-Rideal recombination of oxygen on an RCG surface (which consist of silica by 94%), as given by Eq. (42). In particular, these constraints are caused by significant differences in the materials despite their similar chemical compositions: β-cristobalite illustrates an ideal case of a regular crystal lattice, whereas RCG has an amorphous structure. The probabilities were compared with accuracy to a constant factor used for multiplying the reaction rate constant in the present model. Thus, we compared the general character of the recombination probability dependence on the impinging atom energy rather than its exact values. The results of these comparisons are plotted in Fig. 6. We can conclude that the behavior of the recombination probability as a function of the kinetic energy of the gas-phase atom computed in the present work is in good agreement with the data obtained in *ab initio* calculations for an ideal crystal surface. It should be also noted that the results of probability comparisons offer one more argument for choosing the thresholdless formula of the recombination probability, which was discussed above in Sec. III B.

C. 2D flow around a cylinder with surface catalytic processes

The proposed DSMC surface chemistry models were used for estimating the surface catalycity effects on a reentry

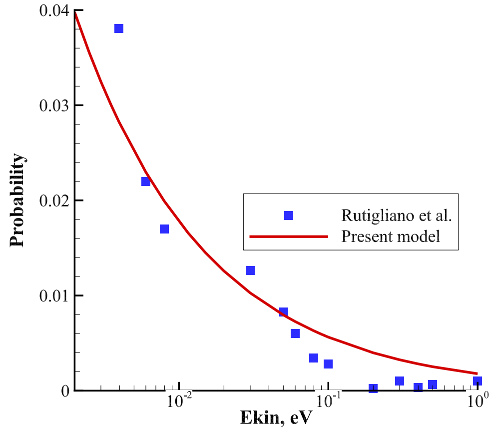


FIG. 6. Comparison of the computed probability of ER-recombination for oxygen on silica with MD calculations of Ref. 60.

vehicle aerothermodynamics. A 2D flow over a cylinder with a radius of 1 m was simulated numerically for the conditions, corresponding to the point of the atmospheric reentry trajectory at the altitude of approximately 85 km (see Table IV). This test case was considered in Ref. 19. The Knudsen number, presented in Table IV, is based on the cylinder radius. The variable soft sphere model of Ref. 61 was used for the elastic collisions. The Larsen—Borgnakke model (Ref. 57) with temperature-dependent vibrational collision number correction of Ref. 62 was employed for rotational-translational and vibrational-translational energy transfer. The kinetic mechanism of Ref. 63, which includes 19 dissociation and exchange reactions for 5 species: N, O, N_2, NO, and O_2 and the total collision energy model (Ref. 2) was used to take into account gas-phase chemical reactions. The gas-phase recombination was neglected which is a reasonable assumption for the low pressures typical of high-altitude reentry flows. More details on the DSMC modeling of elastic and inelastic collisional processes and model parameters can be found in Ref. 19.

The parameters of the DSMC method have been chosen to provide numerically accurate results of the simulation. The chosen time step of 250 ns was significantly lower than the mean collision time in almost entire computational domain. The collisional grid was automatically adapted in the dense regions of the flow to ensure that the flowfield is resolved to the length scales less than the local mean free path. About 2.5×10^6 of simulated molecules and about 400 thousand collisional cells were used in a typical computations. The number of simulated molecules in a virtual volume with the linear size equal to the local mean free path was greater than unity in the whole computational domain which ensures numerically accurate results for the DSMC modeling (Ref. 64). The computations with a larger time step and smaller number of simulated molecules and collisional cells provided results nearly identical to those obtained with the mentioned reference parameters.

TABLE IV. Free-stream conditions. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

Knudsen number	0.01
Velocity V (m/s)	7810
Density ρ (kg/m^3)	5.66×10^{-6}
Flow temperature T_g (K)	178.2
Mole fraction of N_2	0.775
Mole fraction of O_2	0.204
Mole fraction of O	0.021

The general flowfield structure is demonstrated by the translational temperature flowfield presented in Fig. 7. The flow direction is from left to right, and the origin is located at the stagnation point. To assess the effect of surface chemistry, four types of surfaces were considered:

- Non-catalytic surface, where surface recombination does not occur at all;
- Fully catalytic surface, on which all atoms colliding with the surface take part in reactions of recombination (the model Ref. 46 was employed);
- Two types of partially catalytic surfaces, on which surface processes were modeled with the present approach.

The fully catalytic surface was modeled by the simple surface chemistry model (Ref. 46) with a specified constant probability of recombination $\gamma = \gamma_N = \gamma_O = 1$; in this model, the Eley-Rideal and Langmuir mechanism are not distinguished, desorption is not modeled, and the adsorption process is coupled with recombination. The fully catalytic surface, as well as the non-catalytic surface, is taken into account to assess the maximum contribution of surface catalysis to aerothermodynamics.

Two cases of partially catalytic surfaces represent real materials, RCG and α-Al_2O_3, which are usually used in thermal protection systems. They were modeled by the proposed DSMC surface chemistry models based on the macroscopic surface reaction sets of Warnatz *et al.* (RCG-surface, Ref. 33) and Kovalev *et al.* (α-Al_2O_3-surface, Ref. 40).

A diffuse reflection law with full accommodation of energy was used in all four cases for modeling the particle/surface interaction without any surface chemical conversions. Full accommodation of reaction energy was assumed in heterogeneous recombination simulations. The surface temperature was assumed to be equal to 1000 K.

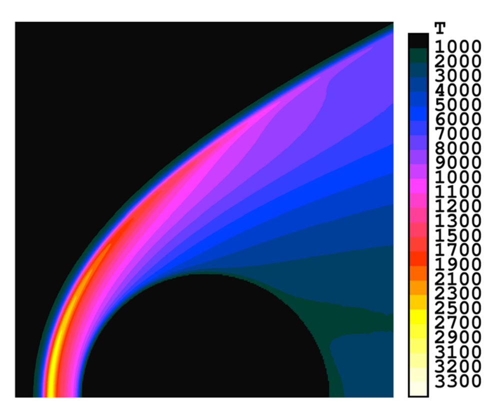


FIG. 7. Temperature flow field. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

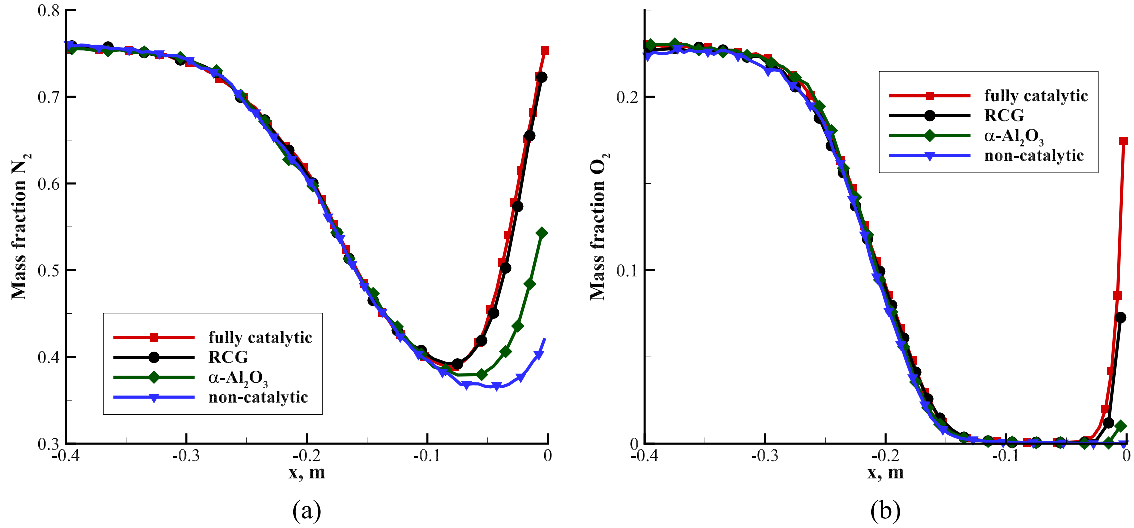


FIG. 8. Mass fraction of the components along the stagnation streamline: (a) N_2; (b) O_2. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

The mass fractions of N_2 and O_2 along the stagnation streamline for all types of surfaces are presented in Figs. 8(a) and 8(b), respectively. The red lines are used in the case of the fully catalytic surface, the blue lines refer to the non-catalytic surface, and the black and green lines illustrate the cases of the RCG and α-Al_2O_3 materials, respectively. The temperature in the shock wave (Fig. 7) increases above 10 000 K, which leads to reactions of dissociation in the gas phase. In turn, dissociation leads to a decrease in temperature behind the bow shock front. It is seen in Fig. 8 that the mass fraction of N_2 decreases due to dissociation almost by 2 times, and the mass fraction of O_2 decreases to zero, i.e., O_2 molecules totally dissociate. For the catalytic cases, the mass fraction starts to increase in the vicinity of the stagnation point because of surface recombination. A small increase in the N_2 mass fraction near the stagnation point can be seen in the non-catalytic case as well. It can be explained by diffusion effects. The mass fraction of N_2 increases almost up to its initial free-stream value in the fully catalytic case, and the plots for the fully catalytic surface and the model for the RCG material are close to each other: the difference between the stagnation point values is about 5%. The plot for the α- Al_2O_3 material is located in the middle between these curves and the plot for the non-catalytic surface. The difference between the plots of the O_2 mass fraction for the fully catalytic case and the models proposed in the present work is greater and reaches 2 times at the stagnation point in the case of the RCG material and 10 times in the case of the α- Al_2O_3 material.

The probabilities of the Eley-Rideal and Langmuir-Hinshelwood recombination reactions (γ_{ER} and γ_{LH}, respectively), as well as the total probabilities of recombination γ calculated with the proposed DSMC models are presented for both N and O components in Table V. Under these free-stream conditions, the total probability of recombination on the RCG material for both components is about 0.1, and the majority of recombination events occur by the Langmuir-Hinshelwood mechanism (about 75% for nitrogen and 83% for oxygen). For the α-Al_2O_3 material, only oxygen recombination is considered, and the total probability of recombination is approximately 4 times lower than that in the case of the RCG surface; almost all recombination events occur by the Eley-Rideal mechanism.

The calculated heat transfer coefficients C_h for all four types of surfaces are presented in Table VI. The table contains the total values of the heat transfer coefficients C_h in the second column, their percentage values relative to the coefficient for the fully catalytic surface $C_{h,FC}$ in the third column, and the percentage values of the surface recombination contribution to the heat flux $C_h - C_{h,FC}$ relative to the maximum possible contribution $C_{h,FC} - C_{h,NC}$, where $C_{h,NC}$ is the heat transfer coefficient for the non-catalytic surface, in the last column (the percentage value of the contribution is 0 for the non-catalytic surface and 100 for the fully catalytic surface). These data also demonstrate the well-known fact (Ref. 4) of the nonlinear dependence of the heat flux on the recombination probability γ: even small values of the surface recombination probability ($\gamma \approx 0.1$ in the considered case) strongly affect the heat flux to the surface—the contribution is 82% of the maximum possible contribution in the RCG model and 25% of the maximum possible contribution in the α- Al_2O_3 model.

Figure 9 shows the heat transfer coefficient distribution over the cylinder surface for all four cases: the fully catalytic case is represented by the red line, the non-catalytic surface is shown by the blue line, and the RCG and α- Al_2O_3 surfaces are

TABLE V. Probabilities of recombination for different DSMC models. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

	RCG		α-Al_2O_3	
Probability of recombination	N	O	N	O
γ	0.118	0.109	0	0.027 572
γ_{LH}	0.080	0.084	0	1.3×10^{-6}
γ_{ER}	0.038	0.025	0	0.027 571

TABLE VI. Heat transfer coefficient. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

Type of surface	C_h	$\frac{C_h}{C_{h,FC}}$ (%)	$\frac{C_h - C_{h,NC}}{C_{h,FC} - C_{h,NC}}$ (%)
Fully catalytic	0.133	100	100
RCG	0.120	90	82
α-Al_2O_3	0.078	59	25
Non-catalytic	0.060	45	0

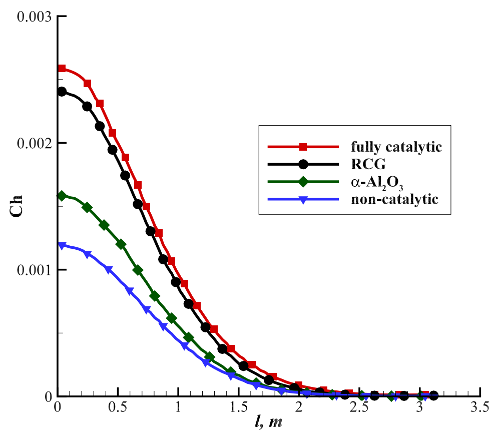


FIG. 9. Heat transfer coefficient along the surface. Reproduced with permission from Molchanova *et al.*, AIP Conf. Proc. **1770**, 040010 (2016). Copyright 2016 AIP Publishing LLC.

illustrated by the black and green lines, respectively (l is the length along the cylinder surface measured from the stagnation point). All graphs have a maximum at the stagnation point and monotonically decrease. It is also seen that the graph for the RCG material is close to the graph for the fully catalytic case, and the graph for the α-Al_2O_3 material is located lower than both of them and closer to the non-catalytic case.

Thus, the surface catalycity models developed on the basis of the proposed approach can be easily implemented in the DSMC code. Verification performed in the present study demonstrates that these models were correctly implemented in the SMILE++ software system. Comparisons with *ab initio* calculations show that the proposed approach adequately predicts the functional dependence of the recombination probability on the incident atom energy for some particular cases. The results of test computations of the flow around a blunted body under high-altitude reentry conditions show that surface catalytic processes can produce a significant effect on the thermal loads on the vehicle surface for TPS materials with low catalytic activity even for sufficiently high Knudsen numbers (Kn ~ 0.01).

VI. CONCLUSIONS

A new approach was proposed for surface chemistry simulations within the framework of the DSMC method. On the one hand, this method takes into account the large variety of surface processes (adsorption, desorption, dissociative adsorption, Eley-Rideal and Langmuir-Hinshelwood recombination); on the other hand, the DSMC method is ideologically close to the phenomenological approach of determining the microscopic properties of molecular interaction on the basis of input macroscopic data on finite rates of the processes, which is used for modeling gas-phase processes in the DSMC method (e.g., in the TCE model).

The approach was applied to two macroscopic mechanisms of surface catalytic processes on bodies in an air mixture: the surface reaction set of Warnatz *et al.* for the RCG material and the surface reaction set of Kovalev *et al.* for the α-Al_2O_3 material. The microscopic parameters of the DSMC model, such as the individual probabilities of the Eley-Rideal recombination and adsorption depending on the surface temperature, the surface density of adsorbed atoms, and the energy of gas-phase particles colliding with the surface, as well as the frequencies of the Langmuir-Hinshelwood recombination and desorption depending only on the surface temperature and surface density of adsorbed atoms, were obtained analytically from given macroscopic reaction rate constants. The developed DSMC models were implemented into the SMILE++ software system, verified by analytical parameters, such as given reaction rate coefficients and analytical solutions for thermal equilibrium conditions, and validated against *ab initio* results; reasonable agreement between the calculations and the *ab initio* data was observed for the energy dependence of the Eley-Rideal reaction probability.

The constructed DSMC surface chemistry models were applied to investigate aerothermodynamics of a 2D cylinder in real flight conditions, taking into account surface catalytic processes. It was found that about 0.1 of all atoms colliding with the RCG surface participate in surface recombination reactions, and most reactions occur by the Langmuir-Hinshelwood mechanism for both nitrogen and oxygen atoms. For the α-Al_2O_3 surface, the probability of recombination is about 4 times smaller, and almost all recombination reactions occur by the Eley-Rideal mechanism. The mass fractions of molecular nitrogen and oxygen were considered. It was noticed that the graphs for the fully catalytic surface and for the RCG model in the case of nitrogen are close to each other, and the difference between them is about 5% at the stagnation point, whereas this difference for the case of oxygen is more discernible and reaches 1.5 times at the stagnation point. The results for the α-Al_2O_3 surface show a lower degree of recombination, and the mass fraction graphs are closer to the non-catalytic case. The total heat flux to the surface was assessed. In the case of the RCG material, the total heat flux is twice greater than the heat flux in the non-catalytic case and is 90% of the heat flux in the case of the fully catalytic surface; however, in the case of the α-Al_2O_3 material, the total heat flux is lower: 59% of the heat flux to the fully catalytic surface. Thus, the computed data show that the contribution of surface catalytic processes to the total heat flux at the surface and the mixture composition is significant even for small values of the recombination probability.

The proposed approach without any significant modifications can be used for constructing a DSMC surface chemistry model on the basis of an arbitrary finite-rate surface reaction set. For this reason, the approach is applicable to all mixtures and surface materials for which surface reaction sets are available in the literature. Though the input reaction set can be obtained from experimental data in the continuum flow regime, the corresponding DSMC surface chemistry model can be also used in strongly rarefied nonequilibrium gas flows.

The approach is sufficiently simple for numerical implementation; in particular, it was formulated as a set of analytical formulas, which are common for various gases and surface materials. Thus, the proposed approach ensures a reasonable compromise between the simplicity, generality, and level of detail necessary for real aerospace applications.

An important feature of the approach should be noted: it allows one to develop DSMC surface chemistry models completely consistent with macroscopic data, which can be used in continuum computations. This feature ensures a meaningful cross-verification between the DSMC and NS codes. Moreover, this consistency gives grounds to extending this approach for applications in DSMC/continuum hybrid codes.

The approach has some constraints because of its phenomenological nature. The degree of reliability of the probability dependences on energy used in this approach can be evaluated only by using comparisons with results of experiments and *ab initio* calculations of such dependences, which are usually unavailable. Therefore, further validation of the approach with the use of the best available data is needed. The future studies should also address the effect of incomplete recombination energy accommodation and application of the proposed approach to macroscopic models taking into account the heterogeneous production of NO, which may have significant effect on the spacecraft aerothermodynamics (see, e.g., Refs. 65 and 66).

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